Mutual Information

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First, an example

Two coin flips $X, Y \in \{0,1\}$, each fair. Work in bits ($\log_2$).

- **Independent.** If the coins are independent, knowing X tells you nothing about Y, and $I(X;Y) = 0$.
- **Perfectly correlated.** If $Y = X$ always, then learning X pins down Y exactly: $I(X;Y) = H(X) = 1$ bit.
- **Noisy channel.** Suppose $P(Y = X) = 0.9$, $P(Y \neq X) = 0.1$. By symmetry Y is still fair, so $H(Y) = 1$. The conditional entropy is the binary entropy of the flip probability, $H(Y \mid X) = H_b(0.1) \approx 0.469$. Hence $I(X;Y) = H(Y) - H(Y \mid X) \approx 1 - 0.469 = 0.531$ bits.

Mutual information interpolates smoothly between zero (independence) and the full entropy of X (functional dependence).

1 Setting

Let X and Y be random variables on a common probability space, taking values in finite (or countable) alphabets $\mathcal{X}$ and $\mathcal{Y}$. Write $p(x,y)$ for the joint probability mass function and $p(x), p(y)$ for the marginals. All logarithms are natural unless otherwise stated; replace log with $\log_2$ to work in bits.

2 Definitions

Definition 1 (Mutual information, KL form)**.** *The* mutual information *of X and Y is*

$$I(X;Y) \;:=\; \mathrm{D_{KL}}\big(p(x,y) \,\|\, p(x)\,p(y)\big) \;=\; \sum_{x\in\mathcal{X}} \sum_{y\in\mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)\,p(y)},$$

Read aloud: *"the mutual information of X and Y is defined as the KL divergence from the joint distribution $p(x,y)$ to the product of marginals $p(x)p(y)$, which equals the double sum over x in $\mathcal{X}$ and y in $\mathcal{Y}$ of p of x,y times log of p of x,y over p of x times p of y." with the conventions $0\log 0 = 0$ and $0\log(0/0) = 0$.*

Definition 2 (Entropic form)**.** *Equivalently,*

$$I(X;Y) \;=\; H(X) - H(X \mid Y) \;=\; H(Y) - H(Y \mid X) \;=\; H(X) + H(Y) - H(X,Y),$$

Read aloud: *"mutual information of X and Y equals entropy of X minus conditional entropy of X given Y, which equals entropy of Y minus conditional entropy of Y given X, which equals entropy of X plus entropy of Y minus joint entropy of X and Y." where $H(X) = -\sum_x p(x)\log p(x)$ and $H(X \mid Y) = -\sum_{x,y} p(x,y)\log p(x \mid y)$.*

3 Main theorem

Theorem 1 (Nonnegativity of mutual information)**.** *For all random variables X, Y,*

$$I(X;Y) \;\geq\; 0,$$

Read aloud: *"the mutual information of X and Y is greater than or equal to zero." with equality if and only if X and Y are independent.*

Proof. By definition, $I(X;Y) = \mathrm{D_{KL}}(p_{XY} \,\|\, p_X p_Y)$, where $p_{XY}(x,y) = p(x,y)$ and $(p_X p_Y)(x,y) = p(x)p(y)$. Both functions are probability mass functions on $\mathcal{X} \times \mathcal{Y}$: the first by construction, and the second because $\sum_{x,y} p(x)p(y) = \big(\sum_x p(x)\big)\big(\sum_y p(y)\big) = 1$.

From the KL-divergence node (concept 38, Gibbs' inequality), for any two pmfs p, q on a common alphabet $\mathcal{Z}$,

$$\mathrm{D_{KL}}(p \,\|\, q) \;=\; \sum_z p(z) \log \frac{p(z)}{q(z)} \;\geq\; 0, \tag{1}$$

Read aloud: "the KL divergence from p to q equals the sum over z of p of z times log of p of z over q of z, and this is greater than or equal to zero." with equality if and only if $p(z) = q(z)$ for every z with $p(z) > 0$ (equivalently, $p = q$ as measures). Applying (1) with $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, $p = p_{XY}$, and $q = p_X p_Y$ gives $I(X;Y) \geq 0$ at once.

For the equality condition: by Gibbs, $I(X;Y) = 0$ iff $p(x,y) = p(x)p(y)$ for every pair $(x,y) \in \mathcal{X} \times \mathcal{Y}$ — which is precisely the definition of independence of X and Y (concept 21). $\square$

Corollary 1 (Symmetry)**.** *$I(X;Y) = I(Y;X)$.*

Proof. The KL form is manifestly symmetric: swapping the summation indices leaves both $p(x,y)$ and $p(x)p(y)$ invariant. Equivalently, $H(X) + H(Y) - H(X,Y)$ is symmetric in X and Y. $\square$

4 Worked example: binary symmetric channel

Let $X \sim \text{Bernoulli}(1/2)$ and $Y = X \oplus N$, with $N \sim \text{Bernoulli}(\varepsilon)$ independent of X, $\varepsilon \in [0,1]$. Then $Y \sim \text{Bernoulli}(1/2)$, so $H(X) = H(Y) = \log 2$, and $H(Y \mid X) = H_2(\varepsilon) := -\varepsilon \log \varepsilon - (1-\varepsilon)\log(1-\varepsilon)$. Therefore

$$I(X;Y) \;=\; \log 2 - H_2(\varepsilon).$$

Read aloud: "the mutual information of X and Y equals log 2 minus the binary entropy of ε." At $\varepsilon \in \{0,1\}$ the channel is deterministic and $I = \log 2$; at $\varepsilon = 1/2$ the channel is pure noise and $I = 0$ — consistent with Theorem 1, since $\varepsilon = 1/2$ makes $X \perp Y$.